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Pre Medicine Association Kurunegala 2024

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය (ආදර්ශ), 2024

General Certificate of Education (Adv. Level) Examination (Model), 2024

රසායන විද්‍යාව I
Chemistry I

09 S I

පැය දෙකයි
Two hours

Important

- Answer all the questions.
- Write your Index number in the space provided in the answer sheet.
- In each of the questions 1 to 50, pick one of the alternatives form (1), (2), (3), (4), (5) which is correct or most appropriate and mark your response on the answer sheet with a cross (x) in accordance with the instructions given on the back of the answer sheet. .

01. Which of the following statements in incorrect?
    (1) According to the Bohr theory proposed by Niels Bohr, when electrons move in a certain energy level they do not emit or absorb energy.
    (2) The $\frac{e}{m}$ ratio obtained for the cathode rays is always greater than the $\frac{e}{m}$ ratio obtained for the positive rays of any gas inside a cathode ray tube.
    (3) In the presence of a magnetic field, α and β rays deviate in opposite directions with equal angles while γ rays shows no deviation.
    (4) Plum pudding model and Golf ball model of an atom are proposed by J. J. Thomson and John Bolton respectively.
    (5) In the gold leaf experiment, a greater proportion of α rays shoe no deviation.

02. During the flame test if an excited mole of atoms emits 180.66 kJ of energy, the wavelength of the emitted electromagnetic radiation is,
    (1) 497.2 nm (2) 662.6 nm (3) 662.6 m (4) 1100.0 nm (5) 6626. 0 nm

03. The increasing order of bond lengths of a, b, c, d, e and f bonds in the following molecule is,

```
            O                NH2     O
            ||               |       ||
    H3 C —— C —— C ≡ C —— C = C —— C —— H
         a     b   c   d    e  | f
                               H
```

    (1) c < e < b = d < f < a (2) c < e < b < d < f < a (3) a < f < d < b < e < c
    (3) a < f < b < d < e < c (5) c < e < a < f < d < b

04. Molecules with same molecular shapes as in ICl_4^-, IBr_2^- and BrO_3^- respectively is,
    (1) ClO_4^-, CS_2, XeO_3 (2) XeF_4, ClO_2^-, XeO_3 (3) XeF_4, XeF_2, XeO_3
    (4) SF_4, ClO_2^-, ClO_3^- (5) SF_4, XeF_2, ClO_3^-

05. What is the IUPAC name of the following compound?

$$CH_3 - CH_2 - \underset{}{\overset{\overset{\displaystyle OH}{|}}{CH}} - \overset{\overset{\displaystyle CHO}{|}}{CH} - CH_2 - CH_3$$

(1) 1–methyl–3–carboxyl pentane (2) 3–carboxy–5–hydroxy hexane
(3) 2–ethyl–4–hydroxy pentanal (4) 3–formyl–5–hydroxy pentane
(5) 2–hydroxy–3–ethyl pentanal

06. The increasing order of the boiling points is,
(1) $NH_3 < CH_4 < HF < H_2O$ (2) $H_2O < HF < NH_3 < CH_4$ (3) $CH_4 < NH_3 < HF < H_2O$
(4) $HF < CH_4 < NH_3 < H_2O$ (5) $NH_3 < HF < H_2O < CH_4$

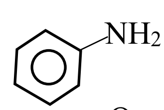
07. A solution is made by adding 250 cm^3 of 0. 15 mol dm^{-3} of 0.15 mol dm^{-3} Na_2SO_4 into 750 cm^3 of 0.10 mol dm^{-3} NaCl solution. The Na composition of the new solution in ppm is,
(1) 3450 (2) 2588 (3) 1725 (4) 3.45 (5) 345

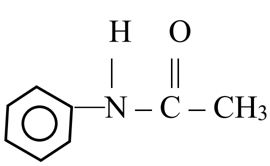
08. Consider the following reaction occuring at 298 K temperature $A_{2(g)} + 3B_{2(g)} \longrightarrow 2AB_{3(g)}$; $\Delta H < 0$
Which of the following statements is correct regarding the above reaction?
(1) At any temperature, ΔG of the above reaction has a negative value.
(2) When the above reaction occurs, the entropy of the surrounding decreases.
(3) At low temperatures, the above reaction is spontaneous.
(4) The entropy of the system increases when the reaction occurs.
(5) At all temperatures, the above reaction in non–spontaneous.

09. Which of the following statements is incorrect?
(1) Radius of N^{3-} anion is greater than that of the O^{2-} anion.
(2) All main group anions have the Nobel gas electron configuration of $ns^2\, np^6$
(3) Some main group cations do not have the Nobel gas configuration.
(4) The highest oxidation number of the elements in the third period of the periodic table increases left to right.
(5) +4 oxidation state becomes more stable when moving downwards along the 16th group.

10. In an acidic medium, the oxidizing agent which is needed in the least amount of moles to oxidize a given KI solution into I_2 is,
(1) $K_2Cr_2O_7$ (2) $KMnO_4$ (3) $FeCl_3$ (4) K_2CrO_4 (5) MnO_2

11. The increasing order of the basicity is,
A - $CH_3CH_2CH_2NH_2$

B - 

C - 

D - $CH_3 - \overset{\overset{\displaystyle O}{\|}}{C} - NH_2$

(1) B < D < C < A (2) A < D < C < B (3) D < B < C < A
(4) B < D < A < C (5) A < D < B < C

12. The following reaction occurs between, the two gases NO and H_2

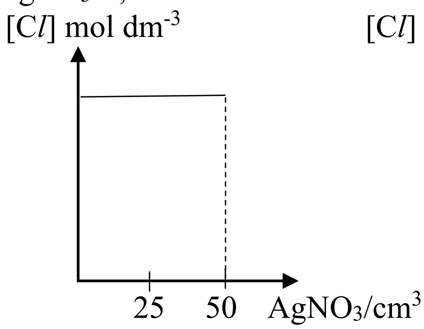
$$2\,No_{(g)} + 2\,H_2O_{(g)} \longrightarrow N_{2(g)} + 2\,H_2O_{(g)}$$

Step I	$NO_{(g)} + NO_{(g)} \rightleftharpoons N_2O_{2\,(g)}$		Fast
Step II	$N_2O_{2(g)} + H_{2(g)} \rightleftharpoons H_2O_{(g)} + N_2O_{(g)}$		Slow
Step III	$N_2O_{(g)} + H_{2(g)} \rightleftharpoons N_{2\,(g)} + H_2\,O_{(g)}$		Fast

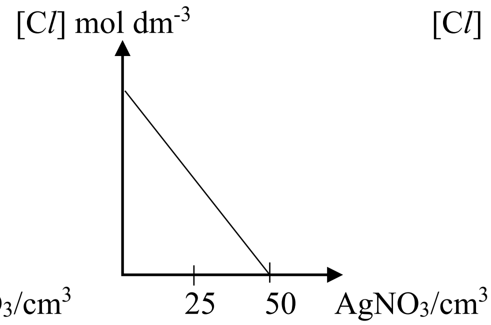
Which of the following statements is correct regarding the above reaction?

(1) H_2 acts as a catalyst.
(2) The rate equation of the above rection is $R = k\,[N_2O_{2(g)}]$ where k is a constant.
(3) Overall order of the reaction is 2.
(4) The rate of the overall reaction can be increased by in increasing the concentration of $NO_{(g)}$
(5) The molecularly of the rate determing step is 1.

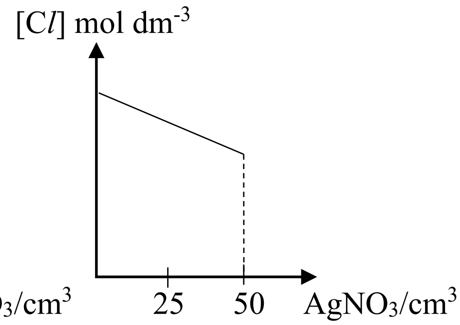
13. 50 cm^3 of 0.1 mol dm^{-3} is gradually added to a 50 cm^3 of 0.1 mol dm^{-3} $BaCl_2$ solution. The graph which accurately depicts the change in the Cl^- concentration with respect to the added volume of $AgNO_3$ is,



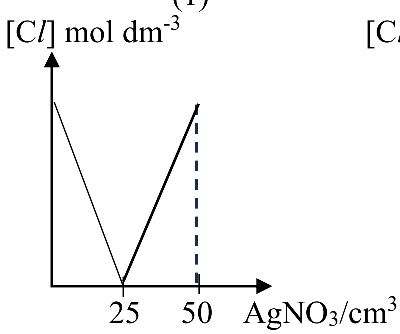
(1)



(2)



(3)



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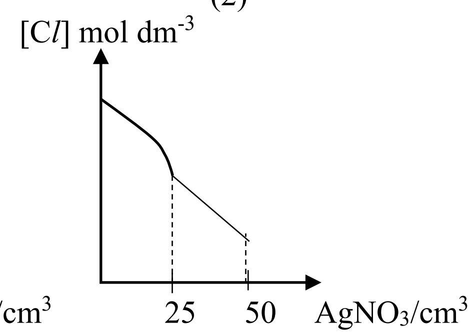


(5)

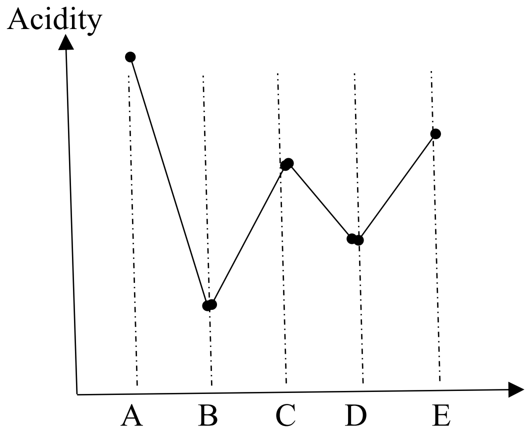
14. The volume (in cm^3) needed from 0.01 mol dm^{-3} $K_2Cr_2O_7$ solution to completely react with 25cm^3 of 0.2 mol dm^{-3} FeI_2 aqueous solution in an acidic medium is,

(1) 8.33 (2) 10.00 (3) 16.67 (4) 50.00 (5) 25.00

15. A right container with a volume of 16.628 dm^3 is sealed after inserting 4.20 g of $N_{2(g)}$, 3.20 g of $O_{2(g)}$ and $X_{(g)}$ of $He_{(g)}$..By using an electronic method, $N_{2(g)}$ and $O_{2(g)}$ are reacted together in the gas mixture. $O_{2(g)}$ is completely consumed by the above. reaction and the only gaseous product forms is $NO_{(g)}$. If the final pressure inside the container is 7.5×10^5 Pa at 27 C^0, the initially inserted mass of $He_{(g)}$ is, (Assume all the gases act as ideal gases under above conditions and molar mass of N =14, O = 16 and He = 4)

(1) 1.00 g (2) 19.0 g (3) 16.3 g (4) 4.75 g (5) 1.90 g

16. Variation of the acidity of four compounds A, B, C, D and E are given below.



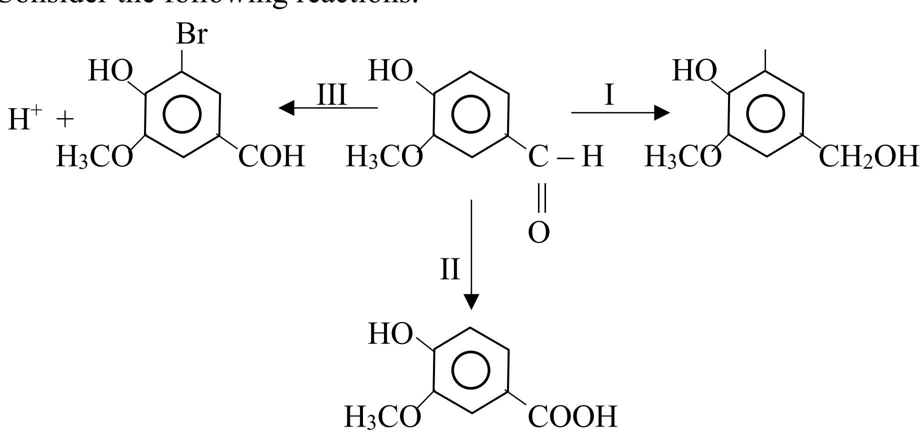
The compounds A, B, C, D and Dand respectively,

(1) Acetic acid, Ethanal, Phenol, Ethyne, Ethene
(2) Acetic acid, Ethene, Ethanal, Ethyne, Phenol
(3) Acetic acid, Ethyne, Phenol, Ethene, Ethanal
(4) Phenol, Ethene, Ethanal, Ethyne, Acetic acid
(5) Acetic acid, Ethyne, Ethanal, Ethene, Phenol

17. The bond dissociation enthalpy of S = O bond is x kJ mol^{-1} and the bond dissociation enthalpy of S – O bond is y kJ mo1^{-1}. The average bond dissociation enthalpy of S – O bond in SO_3^{2-} onion in kJ mol^{-1} is,

(1) $\frac{x+y}{3}$ (2) $\frac{x}{3}$ (3) $\frac{2y+x}{3}$ (4) $\frac{2x+y}{3}$ (5) $\frac{3y+x}{3}$

18. Consider the following reactions.



Types of conversion occur at steps I, II and III are respectively,

(1) Oxidation, Reduction, Addition
(2) Reduction, Oxidation, Addition
(3) Reduction, Oxidation, Emission
(4) Oxidation, Reduction, Substitution
(5) Reduction, Oxidation, Substitution

19. An aqueous solution A, gives a green coloured precipitate when treated with aqueous NaOH. When the above precipitate is heated with acidic $KMnO_4$ a gas evolves giving a solution π. The aqueous solution A also gives a white precipitate in the acidic medium when treated with Ca^{2+} ion solution. The Solution π, when treated with SCN ion solution and NH_4Cl/ NH_4OH separately a dark coloured solution and a brick red precipitate is obtained respectively. The aqueous solution A is,

(1) $FeSO_4$ (2) $NiCl_2$ (3) $CuCO_3$ (4) $FeSO_3$ (5) FeC_2O_4

20. At T temperature, gas A inside a rigid container dissociates as given below. $2A_{(g)} \rightleftharpoons 2B_{(g)} + C_{(g)}$. At temperature T, the pressure inside the container before and after reaching the dynamic equilibrium among gases is P_0 and P_1 respectively. The correct expression for Kp of the above equilibrium system at T temperature is,

(1) $\frac{4(P_1 - P_0)^3}{(3P_0 - 2P_1)^2}$ (2) $\frac{(P_1 - P_0)^3}{4(P_0 - P_1)^2}$ (3) $\frac{4(P_1 - P_0)^3}{(P_1 - 2P_0)^2}$ (4) $\frac{(P_1 - P_0)^3}{(P_1 - 2P_0)^2}$ (5) $\frac{(P_1 - P_0)^3}{(3P_0 - P_1)^2}$

21. A, B and C are three aqueous coropounde. When B is gradually added in excess A, a precipitation in formed and then dissolved. The same observation can be seen when B is added gradually in excess to C aqueous compound.

A, B and C are,

(1) $CuCl_2$, NaOH and $AlCl_3$ (2) $CuCl_2$, NH_4OH and $AlCl_3$

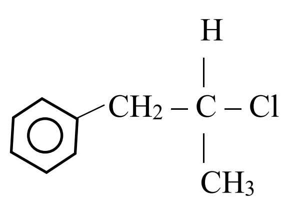
(3) $CrCl_3$, NaOH and $AlCl_3$ (4) $CrCl_3$, $Ba(OH)_2$ and $Al_2(SO_4)_3$

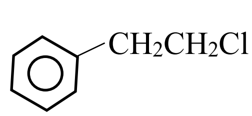
(5) $ZnCl_2$, $Ba(OH)_2$ and $(NH_4)_2CO_3$

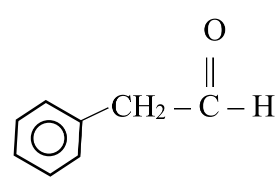
22. When Na_2S salt is completely reacted with V cm^3 of 0.5 mol dm^{-3} HNO_3 solution, $NaNO_2$ is given while emitting SO_2 gas. Then the obtained solution is treated with 80 cm^3 of 0.1 mol dm^{-3} of $KMnO_4$ solution in the acidic medium, turning the purple colour into colourless while emitting no gases. Assuming $SO_{2(g)}$ does not dissolve in the solution, the valve of V is,

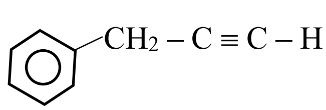
(1) 10 (2) 15 (3) 25 (4) 40 (5) 45

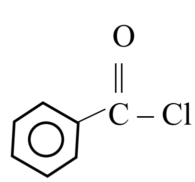
23. Compound Q is given by reacting P with $NaOH_{(ag)}$, while R is given by reacting Q with H^+ / $KMnO_4$. Only the compound R among P, Q, R gives Orange precipitate when treated with 2,4–DNP. The Compound P Could be,

(1) 

(2) 

(3) 

(4) 

(5) 

24. What is the correct expression for the equivalence point pH value in a titration of 25.00 cm^3 of C_O mol dm^{-3} CH_3COOH solution with NaOH at 25 ^0C, if S is the concentration of salt formed, K_a is the acid dissociation constant and K_w is the water dissociation constant at 25 ^0C?

(1) $pH = \frac{1}{2}pK_a + \frac{1}{2}pK_w + \frac{1}{2}\log S$ (2) $pH = pK_w + \log S$

(3) $pH = pK_a$ (4) $pH = \frac{1}{2}pK_a - \frac{1}{2}\log C_0$

(5) $pH = -\frac{1}{2}pK_a - \frac{1}{2}pK_w + \frac{1}{2}\log S$

25. The increasing order of the reducing ability is,

$Fe^{3+}{}_{(aq)} + e \longrightarrow Fe^{2+}{}_{(aq)}$; $E^\theta = +0.77$ V

$Fe^{3+}{}_{(aq)} + 3e \longrightarrow Al_{(s)}$; $E^\theta = -1.66$ V

$Br_{2\,(aq)} + 2e \longrightarrow 2Br^-{}_{(aq)}$; $E^\theta = +1.08$ V

(1) $Br^-{}_{(aq)} < Fe^{2+}{}_{(aq)} < Al_{(s)}$ (2) $Fe^{2+}{}_{(aq)} < Al_{(s)} < Br^-{}_{(aq)}$ (3) $Al_{(s)} < Br^-{}_{(aq)} < Fe^{2+}{}_{(aq)}$

(4) $Al_{(s)} < Fe^{2+}{}_{(aq)} < Br^-{}_{(aq)}$ (5) $Br^-{}_{(aq)} < Al_{(s)} < Fe^{2+}{}_{(aq)}$

26. Some experimental data on the three isomers A, B and C with the molecular formula of $C_5H_{10}O$ are given below.

A - Shows geometric isomerism and able to decolorize Br_2 water.

B - Shows optical isomerism but unable to give orange precipitate when treated with Brady's reagent.

C - Gives silver mirror when tread with Tollen's reagent.

A, B and C are respectively,

(1) $CH_3-CH=\overset{CH_3}{\overset{|}{C}}-CH_2OH,\ CH_3CH_2-\overset{CH_3}{\overset{|}{CH}}-CHO,\ CH_3CH_2CH_2CH_2CHO$

(2) $CH_2=\overset{CH_3}{\overset{|}{C}}-CH_2CH_2OH,\ CH_2=CH-\overset{OH}{\overset{|}{CH}}-CH_2CH_3,\ H-\overset{O}{\overset{||}{C}}-CH_2-\overset{CH_3}{\overset{|}{CH}}-CH_3$

(3) $CH_3\,CH_3-\overset{CH_2OH}{\overset{|}{C}}=CH_2,\ CH_3CH_2-\overset{CH_3}{\overset{|}{CH}}-CHO,\ CH_3CH_2-\overset{O}{\overset{||}{C}}-CH_2CH_3$

(4) $CH_3\,CH=CHCH_2CH_2OH,\ CH_2=CH-\overset{CH_3}{\overset{|}{CH}}-CH_2OH,\ CH_3-\overset{CH_3}{\overset{|}{\underset{\underset{CH_3}{|}}{C}}}-CHO$

(5) $CH_3\,CH_2CH=CHCH_2OH,\ CH_2=CH-\overset{OH}{\overset{|}{CH}}-CH_2CH_3,\ CH_3-\underset{\underset{CH_2OH}{|}}{CH}-CH=CH_2$

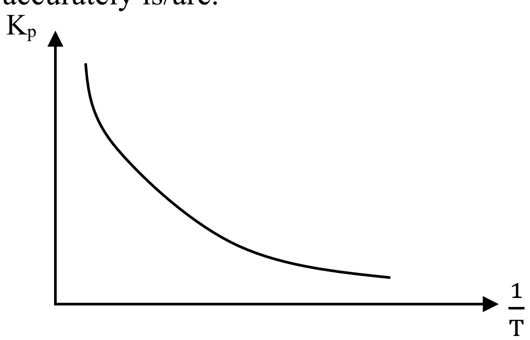
27. In an aqueous solution of Na_2OH and Na_2CO_3, the molar ratio between NaOH : Na_2CO_3 is 1 : 2. When 25.00 cm^3 of this solution is titrated with 0.1 mol dm^{-3} HCl solution, using phenolphthalein as the indicator, at the end point 15 cm^3 of the acid is consumed. If the some titration is done using methyl orange as the only indicator, the end point volume of acid consumed is,
(1) 15.00 cm^3 (2) 20.00 cm^3 (3) 25.00 cm^3 (4) 30.00 cm^3 (5) 30.00 cm^3

28. $a_{(g)} + b_{(g)} \longrightarrow c_{(g)}$
At 298 K, the above reaction is carried out experimentally to obtain following data. Note that the above reaction is not in the balanced form.

Initial concentration of a (mol dm^{-1})	Initial concentration of b (mol dm^{-1})	Initial rate of reaction (mol dm^{-3} min^{-1})
2.5 x 10^{-3}	1.2 x 10^{-3}	1.96 x 10^{-5}
2.5 x 10^{-3}	0.6 x 10^{-3}	9.80 x 10^{-6}
5.0 x 10^{-3}	1.2 x 10^{-3}	7.84 x 10^{-5}

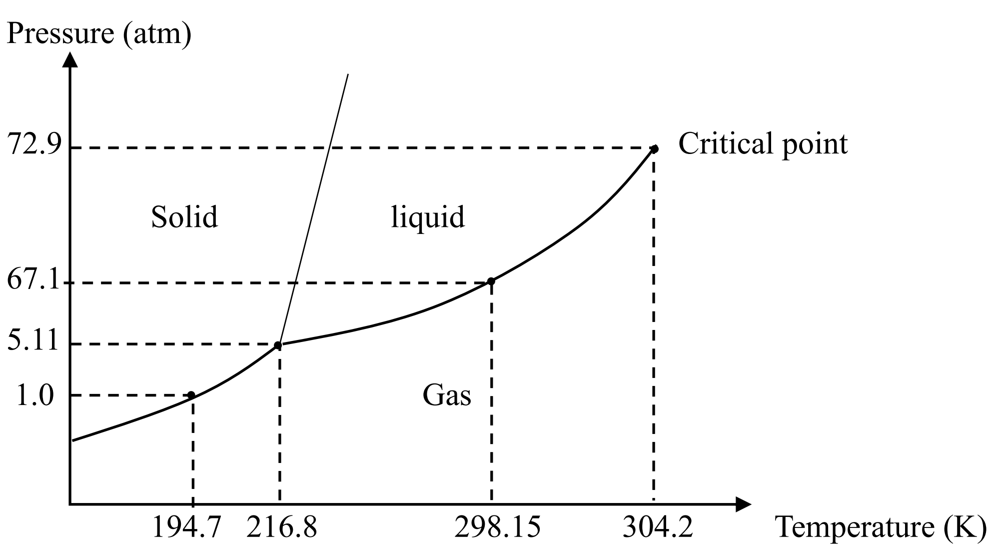
The correct expression regarding the rate of the above reaction is,
(1) rate $\propto$ [a] (2) rate $\propto$ $[a]^2$ (3) rate $\propto$ [b]
(4) rate $\propto$ [a] [b] (5) rate $\propto$ $[a]^2[b]$

29. The graph below shows the variation of the equilibrium constant K_p of the reaction $X_{(g)} + Y_{(g)}$ $Z_{(g)}$ against the reciprocal of the absolute temperature. The conclusions that we can make accurately is/are.



a. Forward reaction is exothermic.
b. At high pressures, the ratio of $Z_{(g)}$ in the mixture increases.
c. At high temperatures, the ratio of $Z_{(g)}$ in the mixture increases.

(1) a only (2) a and c only (3) b only
(4) b and c only (5) all a, b and c

30.


The incorrect statement regarding the above phase diagram of CO_2 is,

(1) At 200k, CO_2 gas calicoes condensation, when increasing the pressure from 1 atm to 50 atm gradually.

(2) At the pressures below 5.11 atm, gaseous CO_2 cannot be converted to its liquid state.

(3) At 298.15 K temperature and 67.0 atm pressure, both gas and liquid CO_2 CO - exist in an dynamic equilibrium.

(4) The boiling point of CO_2 increases with pressure.

(5) At the temperature above 304.2 K, gaseous CO_2 cannot be converted to its liquid state.

- For each of the questions from 31 to 40, One or more responses out of the four responses (a), (b), (c), and (d) given is /are correct. Select the correct response/responses. In accordance with the instructions given, on your answer sheet mark,

(1) If only (a) and (b) are correct (2) if only (b) and (c) are correct.

(2) If only (c) and (d) are correct (4) if only (d) and (a) are correct.

(3) If any other number or combination of responses is correct.

Summary of the above Instructions

(1)	(2)	(3)	(4)	(5)
Only (a) and (b) are correct	Only (b) and (c) are correct	Only (c) and (d) are correct	Only (d) and (a) are correct	Any other numbers combination of responses is correct.

31. Which of the following statements is/are correct regarding the d-block elements and their compounds.

(a) The coordination number of a coordination compound is influenced by the size of the central metal ion of the coordination compound.

(b) The aqueous solutions of d-block ions are colored due to the presence of unpaired electrons in their d – orbitals.

(c) All the d – block elements are metals with relatively higher roughness compared to the metals is s and p blocks.

(d) The colours of aqueous solutions of $[NICl_4]^{2-}$ and $[Cu(NH_3)_4]^{2+}$ complex iron are yellow and blue respectively.

32. The correct statement/statements out of the following is/are.

(a) The TMF value of soap is the mass percentage of free RCOOH in a cake of soap.

(b) In the first step op SMR method of production of H_2 from natural gas, which is needed in production of NH_3, CO gas is reacted with water vapour.

(c) During the Haber process of producing NH_3, due to maintaining a lower temperature the efficiency of the process declines.

(d) When CD gas formed in the lower region of the blast furnance during iron extraction moves up, the thermal stability of CO decreases.

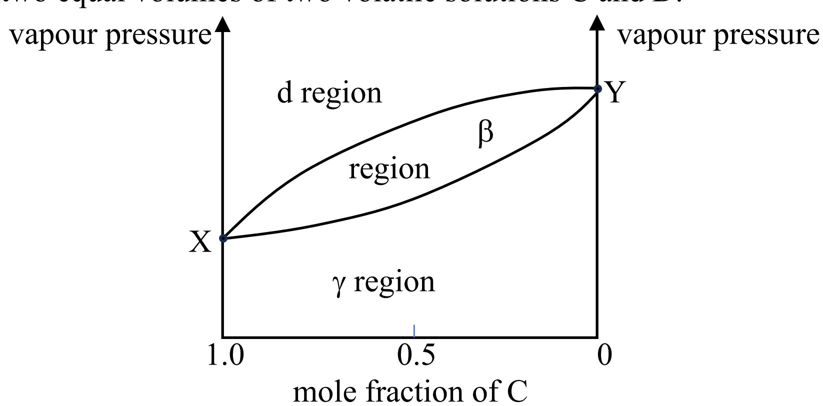
33. A Mg rod a Zn rod which are partially immersed in 1 mol dm^{-3} Mg^{2+} solution and 1 mod dm^{-3} Zn^{2+} solution respectively, are connected by a salt bridge. Using conducting wires the above mentioned Mg and Zn rods are then connected, to two Pt rods 'A' and 'B' respectively, which are also partially immersed in a dilute $CuSO_4$ solution. If the standard reduction potential of $Mg_{(s)}/Mg^{2+}_{(g)}$ is –2.36 V and the standard reduction potential of $Zn_{(s)} / Zn^{2+}_{(aq)}$ is –0.76 V, which of the following statement is/are true regarding the above setup?

(a) The color of the $CuSO_4$ solution gradually becomes lighter with time.
(b) The ammeter reading will take a positive value when the positive terminal and the negative terminal of an ammeter is connected to the Mg rod and 'A' electrode respectively.
(c) After sometime, initially black coloured Pl rod "B" will turn to brick red.
(d) The ratio of the mass change of the Mg rod to its molar mass equals the ratio of the mass change of the Zn rod to its molar mass.

34. The ratio statement/ statements regarding environmental chemistry is/are,

(a) Atmospheric O_2 and water vapour influence the formation of photochemical smog.
(b) HFC used as a substitute for CFC do not contribute to atmospheric pollution.
(c) Most of the gases to the atmosphere are present within the stratosphere.
(d) Although CFCs are present in small amounts in the atm asphere, they have a greater impact on the greenhouse effect compared to many other gases.

35. Given below is the vapor pressure composition diagram for an equilibrium system made by adding two equal volumes of two volatile solutions C and D.

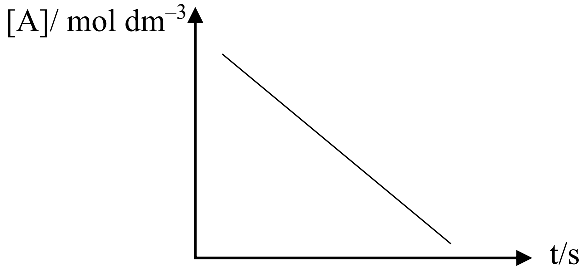


The correct statement /statements regarding the above phase diagram is/ are,

(a) The α region represents the pure liquid phase, the β region denotes the liquid vapour equilibrium phase while the γ region corresponds to the pure vapour phase.
(b) Points X and Y represent the P^0_C and P^0_D respectively.
(c) In the equilibrium system, when the mole fraction of C in the vapour phase is 0.5, the mole fraction of C in the liquid phase is less than 0.5
(d) In the equilibrium system, when the mole fraction of D in the vapour phase is 0.5, the mole fraction of D in the liquid phase is less than 0.5

36. Consider the reaction ; $A + B \longrightarrow$ Products

The following graph depicts the variation of the concentration of A with time, when the concentration of B is in excess and constant: Rate constant $K = 1.5 \times 10^{-2}\ s^{-1}$



The correct statement /statements is/are

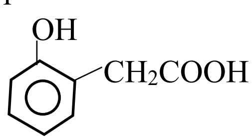
(a) The rate of the reaction decreases with time.

(b) The order of the reaction respect to A is zero.

(c) The half life of the reaction is independent from the concentration of B.

(d) The order of the reaction respect to A is one.

37. When there are two different functional groups in the same carbon chain which can read with each other, cyclic molecules are given by those reactions: Out of the following, which molecules can form cyclic products as mentioned above?

(a) 

and PCl_3

(b) $HO - CH_2CH_2CH_2CH_2NH_2$ and PCC

(c) $H - \overset{\overset{\displaystyle O}{\|}}{C} - CH_2CH_2CH_2 - \overset{\overset{\displaystyle O}{\|}}{C} - NH_2$ and (i) $LiAlH_4$ (ii) H_2O

(d) $Br - CH_2CH_2CH_2 - C \equiv C - H$ and $NaNH_2$

38. Which of the following compounds is/are insoluble in water but soluble in HCl?

(a) $PbCl_2$ (b) BiOCl (c) $Ba(OH)_2$ (d) $CoCl_2$

39. Which of the following statements is/are incorrect?

(a) NH_3 gas is evolved by the reaction between Aniline and HNO_2 occurs of the temperatures greater that 10^0 C

(b) The boiling point of propanol is greater than the boiling point of ethanol.

(c) Grignard regent can acts as both nucleophiles and bases.

(d) The water solubility of 2-butanal is greater than that of 1–butanal.

40. The correct response/responses regarding the iron extraction is/are,

(a) When the temperature increases inside the blast furnance, the spontaneity of the reaction $C_{(s)} + CO_{2\,(g)} \longrightarrow 2CO_{(g)}$ increases.

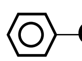
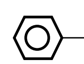
(b) When the temperature increases inside the furnace, the spontaneity of the reaction $2CO_{(g)} + O_{2(g)} \longrightarrow 2CO_{2(g)}$ increases.

(c) As the positive value of ΔG for the reaction $2Fe_{(s)} + O_{2(g)} + \text{heat} \longrightarrow 2FeO_{(s)}$ increases with increasing temperature, the oxidation of iron inside the furnace is avoided.

(d) When moving upwards, in the furnace, the spontaneity of the reaction $2CO_{(g)} + O_{2(g)} \rightarrow 2CO_{2\,(g)}$ increases.

- In the questions from 41 to 50 two statements are given in respect of each question. From the table given below, select the response out of the responses (1), (2), (3), (4) and (5), that best fits the two statements and mark appropriately on your answer sheet.

Response	First statement	Second Statement
(1)	True	True, and correctly explains the first statement
(2)	True	True, but does not explain the first statement correctly.
(3)	True	False
(4)	False	True
(3)	False	False

No.	First Statement	Second Statement
41	When a few drops of $AgNO_3$ is added to an equilibrium system containing $[CoCl_4]^{2-}_{(aq)}$ and $[CoCH_3O)6]^{2+}_{(aq)}$. the solution turns pink.	The equilibrium shifts forward in the reaction, $[CoCl_4]^{2-}_{(aq)} + 6H_2O_{(l)} \rightleftharpoons [CoCH_2O)_6]^{2+}_{(aq)} + 4Cl^-_{(aq)}$ when $Cl^-_{(aq)}$ ion concentration decreases.
42	Tollens reagent can be used to identify $H-\overset{\overset{O}{\Vert}}{C}-H$ from $H-\overset{\overset{O}{\Vert}}{C}-OH$	$H-\overset{\overset{O}{\Vert}}{C}-H$ is oxidized by the Tollens reagent.
43	All the beginning of H-Br addition reaction to propene, the following step occurs $CH_3-CH=CH_2 \quad H-Br$	The π election cloud in the alkene can attach a species with the ability to accept a pair electron to itself.
44	$-CH_2Cl$ with aqueous A_gNO_3 forms a precipitate.	$-C^+H_2$ is highly stable
45	ΔG has a highly negative value in a galvanized cell.	Electrical energy is generated in a galvanized cell by a spontaneous redox reaction.
46	When an inert gas is inserted into a constant volume closed system at equilibrium with $A_{(g)} + 2B_{(g)} \rightleftharpoons 2C_{(g)}$ the system does not change.	After inserting an insert gas to a wnstant volume closed system at equilibrium with $A_{(g)} + 2\,B_{(g)} \rightleftharpoons 2C_{(g)}$, the partial pressures of $A_{(g)}$, $B_{(g)}$ and $C_{(g)}$ does not change.
47	Nernst distribution law cannot be applied to a system in which HCl is distributed in $CHCl_3$ and H_2O.	If the solute concentrations in two immiscible basic solvents are at very lower, values, Nernst law cannot be applied.

48	A mixture of Acetone and $CHCl_3$ forms on ideal solution with a negative deviation.	H bonds are formed when Acetone is mixed with $CHCl_3$
49	In the fractional distillation process of ethanol production, the first distillate is often discorded.	Methanol is a toxic alcoholic compound.
50	In the iron extraction process, at temperatures below 1000 ^{0}C inside the furnace, Fe_2O_3 is reduced by $CO_{(g)}$	With the decreasing temperature, thermal stability of $CO_{2(g)}$ increases than the thermal stability of $CO_{(g)}$

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Pre Medicine Association Kurunegala 2024

අධ්‍යයන පොදු සහතික පත්‍ර (උසස් පෙළ) විභාගය (ආදර්ශ), 2024

General Certificate of Education (Adv. Level) Examination (Model), 2024

රසායන විද්‍යාව II **Chemistry II**	**09 S II**		**පැය තුනයි** ***Three hours***

PART A – STRUCTURE ESSAY

01. (a) Arrange the following in the ascending order according the property given in the backets.

(i) CO_2 , CO_3^{2-}, CH_4 (Electronegativity of C atom)

………….. < …………..< …………

(ii) NO, NO_2, NO_3^- (N - O bond length)

………….. < …………..< …………

(iii) Li_2SO_4, Na_2SO_4, K_2SO_4 (Ionic Character)

………….. < …………..< …………

(iv) NaCl, H_2O, He, H_2Se (Boiling point)

………….. < …………..< ………… < …………

(v) NaCl, $MgCl_2$, $AlCl_3$ (Solubility in water)

………….. < …………..< …………

(vi) $BeCl_2$, OCI_2, NCI_3, CC14 (Band angle)

………….. < …………..< ………… < …………..

(b) (i) Name the ion SCN^-

(ii) Draw the lewis structure of SCN^- ion

(iii) State the oxidation number and the configuration of each atom.

Atom	S	C	N
Oxidation No			
Configuration No			

(iv) Draw three Recessive structure of the above ion.

(c) H_3C_4NS shows acidic characters. In water, this produces anions by releasing H^+ ions. The skeleton of that anion is given below.

```
            H   O
            |   |
    H – O – N – S – O
                |
                O
```

Draw the acceptable lewis structure for the $H_2C_4NS^-$ anion answer the following questions.

(i) Draw the lewis structure

(ii) Complete the following table by considering the atoms N, S and O bonded with H

	O bonded with H	N	S
VSEPR pairs			
Electron pair geometry			
Shape			
Hybridization			
Band angle			

(iii) State the atomic/ hybridized orbitals involved in the formation of bonds in the above lewis structure.

I. O – N   O   N

II. N – S   N   S

III. S – O   S   O

(d) Element A makes A^+ ions. Element P makes P^+ ions. Element B makes B^+ ions. Radius at P^+ ion is greater than the radius of A+ ion.

(i) Which ion has the higher polarizing power from A^+ and P^+ ions.

(ii) Explain the Reason.

(iii) Which compound is more ionic among AB and PB.

02. (a) A is an element which reacts with water and bases to undergo both oxidation and reduction simultaneously.
A produces D, E, F by reacting with cold dilute NaOH
E shows bleaching properties and contains an ion which undergoes disproportionation reaction.
B is a non–metal which belongs to P. block.
B reacts with hot concentrated H_2SO_4 and gives G which is a colorless water soluble gas. In acidic medium, G converts the purple colour $KMnO_4$ solution to a clear colorless solution.
C is a covalent compound made up of A and B. With the reaction of water C gives H which is a strong acid and B precipitate.

(i) Identity the species A - H

A -	E -
B -	F -
C -	G -
D -	H -

(ii) Write the balanced equations for following reactions.

(I) A + Cold. dil NaOH

(II) G + acidic $KMnO_4$ (aq)

(III) Deprotonation reaction of the anion in E.

(IV) C with water

(V) B + hot cone. H_2SO_4

(b) Write the chemical formulae of P, Q, R, S (Balanced equations are required)

(i) P is a water insoluble salt. This salt is made up of an alkali metal and two elements in the p block.
A triatomic gas is released when HCl solution is added to this salt.

(ii) Q is a water soluble salt. This gives lilac colour when subjected to flame test. When BaCl, solution is added to this salt, this gives a yellow precipitate.

(iii) When R solution is mixed with NaCl Solution, it gives a yellow precipitate which is soluble in dilute Ammonia. When R solution is heated black precipitate is obtained.

(iv) The metal in the S salt belongs to 3rd period of the periodic table. This metal reacts with a diatomic gas under a high temperature and produces the salt S. The gas released with the addition of water to this salt, produces a white smoke in the presence of HCl. Except for the gas, it produces a non–gelatinous white precipitin.

(c) Write the balanced chemical equations for following reactions and state the oxidant and the reductant.

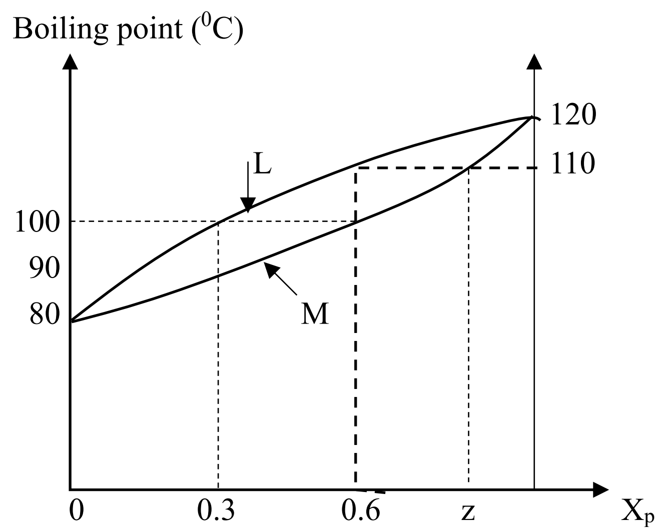
(i) Mg (s) + Conc. HNO_3

Oxidant :- Reductant :-

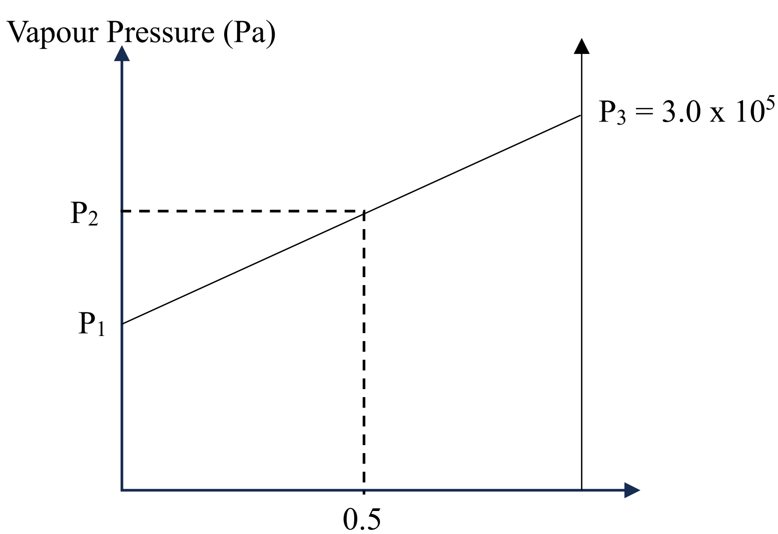
(ii) Cu (S) + Conc. HNO_3

Oxidant :- Reductant :-

03.



(1)



(2)

Two graphs drawn for a mixture which contains ideal P and Q liquids are given above.

(i) Identify the curves L and M.

L –

M –

(ii) Identify graph (1)

(iii) Show that the mole fraction of P_n, Vp, $= \frac{P_P^0}{P_P^0 + P_Q^0}$ for the solution when $X_p = 0.5$

(iv) What is the concentration of the solution obtained when the vapour phose which is at equilibrium with a 100 ^{0}C boiling solution, is liquified completely.

(v) The x axis of graph (2) indicates the mole fraction of a compound. What is that compound? Explain.

(vi) Find P_1 and P_2 if the mole fraction of Q in vapour phase is 0.6 when $X_p = 0.5$

(vii) Explain how to obtain pure Q from a mixture where $X_p = Z$

(b) The weak monobasic organic acid HA dissolves in benzene four times that water. When 80 g of HA is mixed with 50 cm^3 of benzene and 100 cm^3 of water to reach equilibrium, 1 g or HA was found in 10 cm^3 or benzene layer. (HA does not disintegrate in organic lay)

(i) Find the free HA mass in the aqueous layer.

(ii) Find the disintegrated HA mass in the aqueous layer.

04. (a) A, B, C, D are the isomers of the atomic structure C_5H_{10}. A shows geometrical isomerism. B, C, D reacts with HB in the peroxide medium, and the products obtained is mixed with dil. NaOH and those products are reacted with H^+/ $KMnO_4$ E, F, G are obtained respectively. Only F gave a coloured precipitate with 2,4 – DNP. When E and G are mixed with $LiAlH_4$ and the H^+/ H_2O, the obtained products are heated with conc. H_2SO_4 and the dil. H_2SO_4 is added, the product from G give an immediate solution with Lucas reagent and it is H. B is a positional isomer 01 D.

(i) Draw the structures of A, B, C, D, E, F, G and H

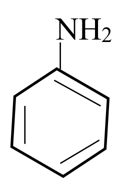
(A)	(B)

(C) (D)

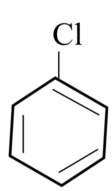
(E) (F)

(G) (H)

(b) Draw the structure of main products I, J, K, L and M in the given reactions.

(i)  $\xrightarrow[(0\,^{0}C - 5\,^{0}C)]{NaNO_2 \text{ / dil. } HCl}$ (I)

(ii) $CH_3 - C \equiv C - H \xrightarrow[\Delta]{HgSO_4 \text{ / dil. } H_2SO_4}$ (J)

(iii)  $\xrightarrow[CH_3 - CH_2 - Cl]{\text{Anhy. } AlCl_3}$ (K)

(iv) $CH_3 - \overset{\overset{\Large O}{\|}}{C} - O - CH_2 - CH_3 \xrightarrow[\text{(ii) } H^{+} / H_2O]{\text{(i) } CH_3 - CH_2 - MgCl}$ (L)

(iv) $CH_3-\underset{\displaystyle H}{\overset{\displaystyle CH_3}{\underset{|}{\overset{|}{C}}}}-\overset{\displaystyle O}{\overset{\|}{C}}-H \xrightarrow{CH_3-\overset{\displaystyle NH_2}{\overset{|}{C}H}-CH_3}$ (M) $\boxed{}$

(c) When the above F compound reacts with 2, 4 – DNP

(i) Write the main reaction type

(ii) Identify the nucleophile (Draw the structure)

(iii) Write the mechanism.

PART B – ESSAY

05. (a) Al 27 ^0C, 0.5 mol of $A_{(s)}$ and 0.8 mol of $B_{(g)}$ is mixed inside a closed rigid container with a volume of 4.157 dm^3. No reaction occur between $A_{(s)}$ and $B_{(g)}$ at 27 ^0C, but when heated upto 127 ^0C $A_{(s)}$ and $B_{(g)}$ react, forming $C_{(g)}$ and comes to equilibrium,

$$2A_{(s)} + 3B_{(g)} \rightleftharpoons 2C(g) \ \ldots\ (1)$$

The formed $C_{(g)}$ moles at this moment is 0.2 moles.

When the above system is heated up to 427 ^0C, in addition to the above mentioned equilibrium another equilibrium occurs by the dissociation of $C_{(g)}$ Inlo $D_{(g)}$ and $E_{(g)}$ as given below.

$$C_{(g)} \rightleftharpoons D_{(g)} + E_{(g)} \ \ldots\ldots\ (2)$$

At this moment, the system contains 0.2 mol of $B_{(g)}$ and 0.25 mol of $D_{(g)}$

(i) Find the total pressure of the equilibrium system at 127 ^0C

(ii) Calculate the equilibrium constant K_p of the equilibrium system (1) at 127 ^0C

(iii) Using the K_p calculated to above part (ii), find the value of K_c at 127 ^0C

(iv) Calculate the partial pressures of each gas in the equilibrium system at 427 ^0C

(v) Calculate equilibrium constants $K_{p(1)}$ and $K_{p(2)}$ for the two equilibrium (1) and (2) respectively at 427 ^0C

(vi) Using the values obtained for K_p and $K_{p(1)}$ at two different temperatures, explain whether the reaction (1) is endothermic or exothermic, giving reason.

(vii) Using a reasonable calculation, show in which direction the equilibrium (1) will shift, if 0.2 mol of $B_{(g)}$ and 0.1 mol of $C_{(g)}$ are externally added to the equilibrium system at 127 ^0C.

(b) $2\,X_{(s)} + Fe_2O_{3(s)} \longrightarrow X_2O_{3(s)} + 2Fe_{(s)}$. Here λ is a p-block compound. Which gives H_2 gas when reacted with aqueous NaOH and with aqueous HCl Acid. The standard formation enthalpies of $\lambda_2O_{3(s)}$ and $Fe_2O_{3(s)}$ respectively at 25 ^{0}C are –1675 kJ mol^{-1} and –825 kJ mol.

(i) Find the element λ

(ii) Calculate ΔH of the above reaction given, at 25 ^{0}C

(iii) At 25 ^{0}C, the standard entropy change ΔS for the given reaction is 15.2 J K^{-1} mol^{-1} Calculate ΔG value for the reaction at the some temperature.

(iv) Calculate the λ moles needed in fusion of 5.20 g of chromium metal at its melting point (Fusion enthalpy of Cress is 14.6 kJ mol^{-1}, relative atomic mass of Cr = 52)

(c) A gaseous product P produced in a certain reaction is absorbed into a solid compound named Q_0 giving $Ca(OH)_{(s)}$ as the only product. When the entire mass of $Ca(OH)_{2(s)}$ obtained is added to water and dissolved, at T temperature a 500 cm^3 of solurated $Ca(OH)_{2(aq)}$ solution with 0.89 g undissolved $Ca(OH)_{(s)}$ precipitate is obtained (AT T temperature, K_{SP} $Ca(OH)_2 = 10.8 \times 10^{-5}$ mol dm^{-3}, K_{SP} $Mg(OH)_2 = 3.2 \times 10^{-11}$ mol dm^{-3})

(i) Identify P and Q.

(ii) Write down the balanced chemical equation for obtaining $Ca(OH)_2$ produce

(iii) Calculate the solubility (x) of $Ca(OH)_2$ in the solution at T temperature.

(iv) Calculate the entire mass of $Ca(OH)_2$ produced in the absorption of P by Q

(v) When adding highly diluted $MgCl_2$ solution into the above saturated $Ca(OH)_{2(aq)}$ solution in a drop by drop procedure, precipitation of $Mg(OH)_2$ is initiated.

I. Calculate the solubility (y) of $Mg(OH)_2$ in pure water at T temperature.

II. Calculate the solubility (z) of $Mg(OH)_2$ in the above $Ca(OH)_2$ solution, at the moment when precipitation of $Mg(OH)_2$ begins.

III. Does the solubility of $Mg(OH)_2$ increase at the moment stated in II above, relative to its solubility in pure water? Briefly explain the reason for your answer.

06. (a) Consider the equation $Na_2S_2O_3 + 2\,HNO_3 \longrightarrow 2\,NaNO_3 + S + SO_2$

The time taken to produce constant S amount was measured in order to find me rate equation.

Reach No.	$Na_2S_2O_3$ Volume (cm^3)	HNO_3 Volume (cm^3)	H_2O Volume (cm^3)	Time (S)
1	25.2	10	14.8	10
2	20	10	20	20
3	25.2	20	4.8	5
4	10	20	20	t

(i) How to measure the time taken to produce constant S amount?

(ii) The volume of water added in each experiment was deferrer what is the reason for that?

(iii) Rate equation cannot be written according to the stoichiometry ratio to the reaction. Explain the reason.

(iv) The rate of the reaction can be indicated as $R \propto [Na_2S_2O_3]^x\,[HNO_3]^y$ Find the values of x and y then calculate the order of the reaction.

(v) Rate of reaction increases with increasing temperature. Write 2 main reason for this.

(vi) Calculate t in the above table.

(b) A 25 cm^3 volume of strong base BOH whose pH is 13. 01 was titrated with 0.25 mol dm^{-3} HCOOH solution placed in biurette. (At this temperature $K_{a\ (HCOOH)} = 1.8 \times 10^{-4}$ mol dm^{-3} and $K_{w\ (HLCOH)} = 1 \times 10^{-14}$ $mol^2\ dm^{-6}$)

(i) What is the initial BOH concentration in the titration flask.

(ii) When 15 cm^3 HCOOH and is added to the flask what is the pH value of the solution.

(iii) What is the volume of the weak acid HIDOH should be added to reach the equilibrium point ?

(iv) What is the pH value at equilibrium point?

(v) What is the pH value of the solution in the flask, when to cm^3 of the acid is added after reaching the equilibrium.

(vi) Using the above calculated pH values, draw the pH graph between the pH value of the solution in the titration flask and the added volume of the acid.

(vii) Shade the region of the pH which shows the buffer action of the solution.

(c) An ideal binary mixture of $A_{(l)}$ and $B_{(l)}$ in a closed system at 27 ^0C, are at equilibrium with their vapour.

(i) Write the equation related to Roult law and identify the terms.

(ii) The total pressure of the equilibrium system made by mixing certain amount of $A_{(l)}$ and $B_{(l)}$ at 27 ^0C is 9×10^4 Nm^{-2} ($P_T = 9 \times 10^4$ Nm^{-2}). At that temperature $P_A{}^0 = 4 \times 10^4$ Nm^{-2} and partial pressure at B, $P_B = 7 \times 10^4$ Nm^{-2}. Volume of vapour phase is 4.157 dm^3. Mole ration of liquid and vapor phase is 9 : 2

I) Find P_B^0

II) Find the total no. of moles in the vapour phase.

III) Find the total no. of moles in the liquid phase find the initial A and B moles.

07. (a) (i) At 25 ^0C an electrochemical cell was made by using standard hydrogen electrode and silver /silver chloride electrode using HCl as the common electrolyte for both electrodes.

$Ag_{(s)}, AgCl_{(s)} / Cl^-_{(aq)}$ $\quad E^0 = 0.22$ V

$Pt_{(s)} / H_2(g) / H^+(aq)$ $\quad E^0 = 0.00$ V

(1) Write the oxidation halfs reaction and reduction half reaction of the above cell.

(2) Write the complete cell reaction.

(3) Indicate the above cell in standard cell notation.

(4) Does the electromotive force of above cell depends on the Cl^- concentration? explain the reason.

(ii) When Cu metal is refined commercially, $CuSO_4$ solution is electrolyzed using Cu electrodes.

(I) State separately what Cu electrodes are used as the anode and the cathode.

(II) Mass of a electron was increased by 31.75 g when electrolysis was done under constant current for 10 minutes.

(a) Write the half reactions for the electrodes.

(b) Calculate the constant current used in the electrolytic reaction.

(b) X and Y are two elements that belong to d block when excess NaOH and A compound is added to a X^{3+} solution which is a ration of X, Yellow colour compound M is formed.

Y^{n+} which is a cation of Y, reacts with the oxoanion N which has the +6 oxidation state of X, oxidizes to X^{3+} only is acidic medium.

(i) Identity X and Y

(ii) Write the chemical formular of compounds A, M and N.

(iii) What is the value of n in Y^{n+}?

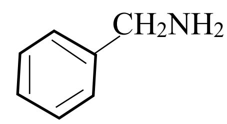
(iv) Write the balanced ionic equation for the reaction between N and Y^{n+}.

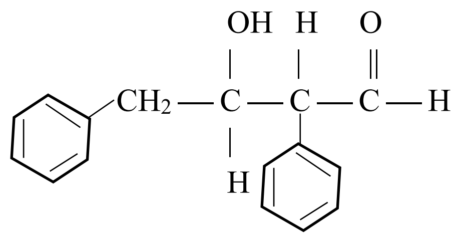
(v) When the reagen C was added to a Y^{n+} solution, a dark blue precipitate was obtained. Write the chemical formula of C.

(vi) Write four oxides 01 x (chemical formula) and state their acidic /basic nature.

(vii) Write the balanced ionic equation of the reaction between X and conc. HCI.

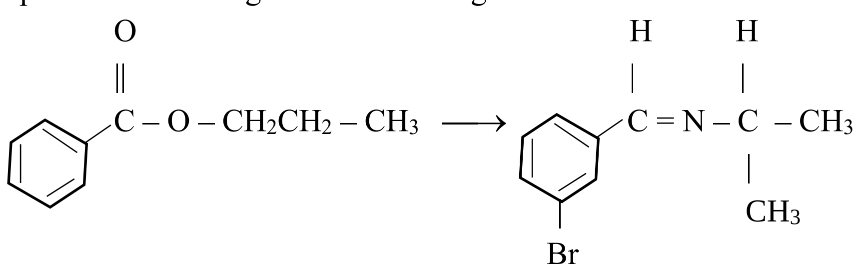
(viii) What is the reason for the oxoanion of X at +6 oxidation state to show colours, even though it does not contain partially filed d orbitales in the + 6 oxidation state of X.

08. (a) Show are way of obtaing x in less than 8 steps, by using inhal compound as 

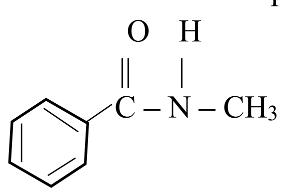


Compound (X)

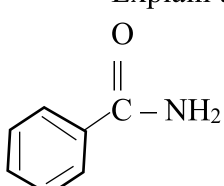
(b) (i) Complete the following conversion using suitable methods.

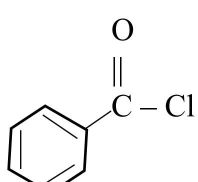


(ii) What is most suitable pair of compound's needed to mark the compound.



Explain the reason.


(A)


(B)

CH_3Cl
(C)

CH_3NH_2
(D)

O

(c) Give the product and the mechanism of the reaction be times aqueous NaOH and $CH_3 - \overset{\|}{C} - H$ (ethanol)

09. (a) In the solution Z, there are 2 cations and 3 anions. When dil NaOH is added to Z solution a green precipitate mixture and a colorless y solution are given. When this precipitate mixture is separated and treated with excess NaOH, Opart of the precipitate dissolves forming a green colour solution B. The B solution and the remaining precipitate is separated ; then 6% H_2O_2 are added respectively to the B solution giving a yellow colour solution C and to the precipitate giving a precipitate called D. When the C solution is treated with AgNO3, a red precipitate E is obtained. A part of F yellow colored solution which is obtained by adding dil. HCl to the formed D precipitate, is treated further with excess dill. HCl to give another yellow coloured solution G. The remaining part of the F solution gives a red coloured solution H when Ammonium Thiocyanate is added separately. The initially obtained y solution is treated with $BaCl_2$ after separating it from the mixture A , and then it gives a white precipitate with a colourless solution. When the solution and the precipitate is separated and both are treated with all HCl separately, a brown gas I is evolved from the solution while a port of the precipitate dissolves giving a colourless solution and a colourless gas J. The remaining part of he precipitate which does not undergo any transformation is L. when the gas J is bubbled through a orange coloured Solution K; which can be made by adding dill. HCl into the earlier formed C solution, the solution K turns into a clear green colour.

(i) Identify all the anions and cations in the Z solution.
(ii) Name the 2 compounds present in A mixture.
(iii) Identify all the compound from B to L. (Chemical Bimulae is expected.)
(iv) Write the balanced chemical equitation for the reaction occur when J gas is bubbled thought K Solution.

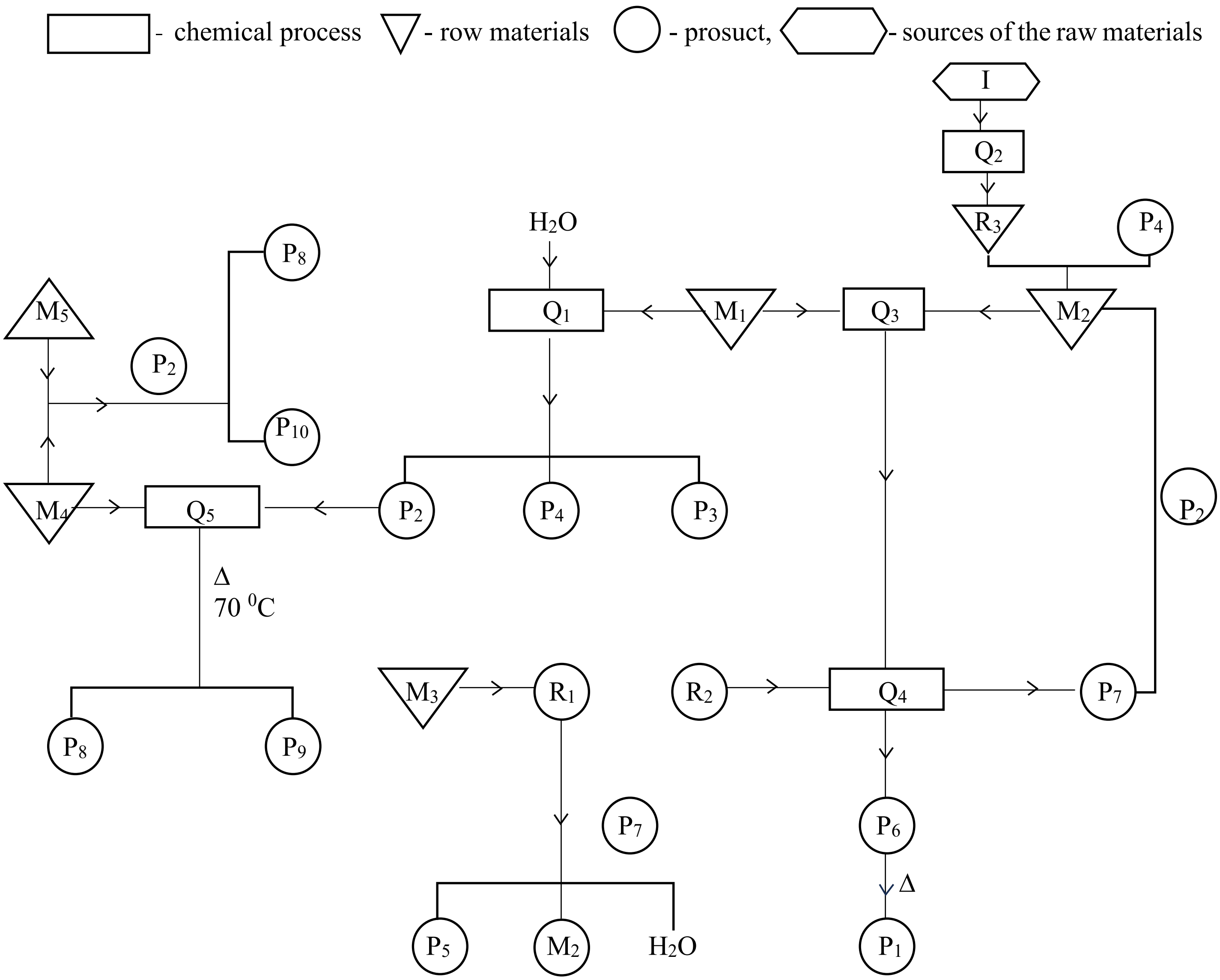
(b) 30 cm^3 of 0.4 mol dm^{-3} $KMnO_4$ solution is needed for the titration of 30 cm^3 of a solution containing both SO_3^{2-} and $C_2O_4^{2-}$ ions. 25 cm^3 of the solution obtained after the titration is then heated with excess $BaCl_2$ to obtain a while precipitate. The dry mass of the precipitate obtained is l.165 g. Assume the density of the solution is 1 g cm^{-3} (Ba-131, O-16, S-32, C-12)

(i) Write balance chemical equations for all the above reactions accrued.
(ii) Find the SO_3^{2-} and $C_2O_4^{2-}$ concentrations in the initials solution separately.
(iii) Calculate the mass percentage of carbon in the initial solution.

10. (a) The following flow chart shows the production process of three important componds P_1, P_2, M_2 and some other compounds P_3, P_4, P_5, P_6, P_7, P_8, P_9 and P_{10} which are derived by those 3 mail compounds. Out of the above,

- P_4 is used in weather forecast.
- P_7 can be used to regenerate M_2

- P_3 reacted with water gives a solution with bleaching property.
- P_8 is used in cosmetic7.



(i) Identify P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8, P_9 and P_{10}

(ii) Identify M_1, M_2, M_3, M_4, M_5 and R_1, R_2, R_3, I

(iii) Name the process Q_1, Q_2,Q_3, Q_4 and Q_5

(iv) Draw a labeled diagram of the apparatus used in process Q_1

(v) Using specific chemical equations briefly explain the process of Q_3 and Q_4

(b) (i) In the making of essential oils cinnamon is frequently used. Write one example each, for the chemical compounds extracted from leaves, barks and roots of cinnamon.

(ii) Write 3 uses of essential oils.

(iii) Write down 2 advantage of steam distillation method.

(iv) Explain the statement "The boiling point of the mixture is lower than the boiling point of pure water and essential oil".

(v) Write down the balanced chemical reactions related to the process of natural vinegar production.

(c) (i) What is defines as Eutrophication?

(ii) Write down the two main ions cause eutrophication and the ways they are added to water.

(iii) A water sample from an atrophied water body is collected to a reagent bottle and then 1 cm^3 of $MnSO_4$ and 2 cm^3 alkaline KI solution are added and mixed thoroughly to give a solution with a precipitate. Then conc H_2SO4 added to the solution dissolving the precipitate and then 50 cm^3 of the obtained solution is titrated with a 0.02 mol dm^3 $Na_2S_2O_3$ consumed, find the amount of dissolved O_2 in the water sample in ppm.

(iv) Write down 3 water quality parameters that change due to acid rains and 2 impacts on aquatic organisms due to those charges.